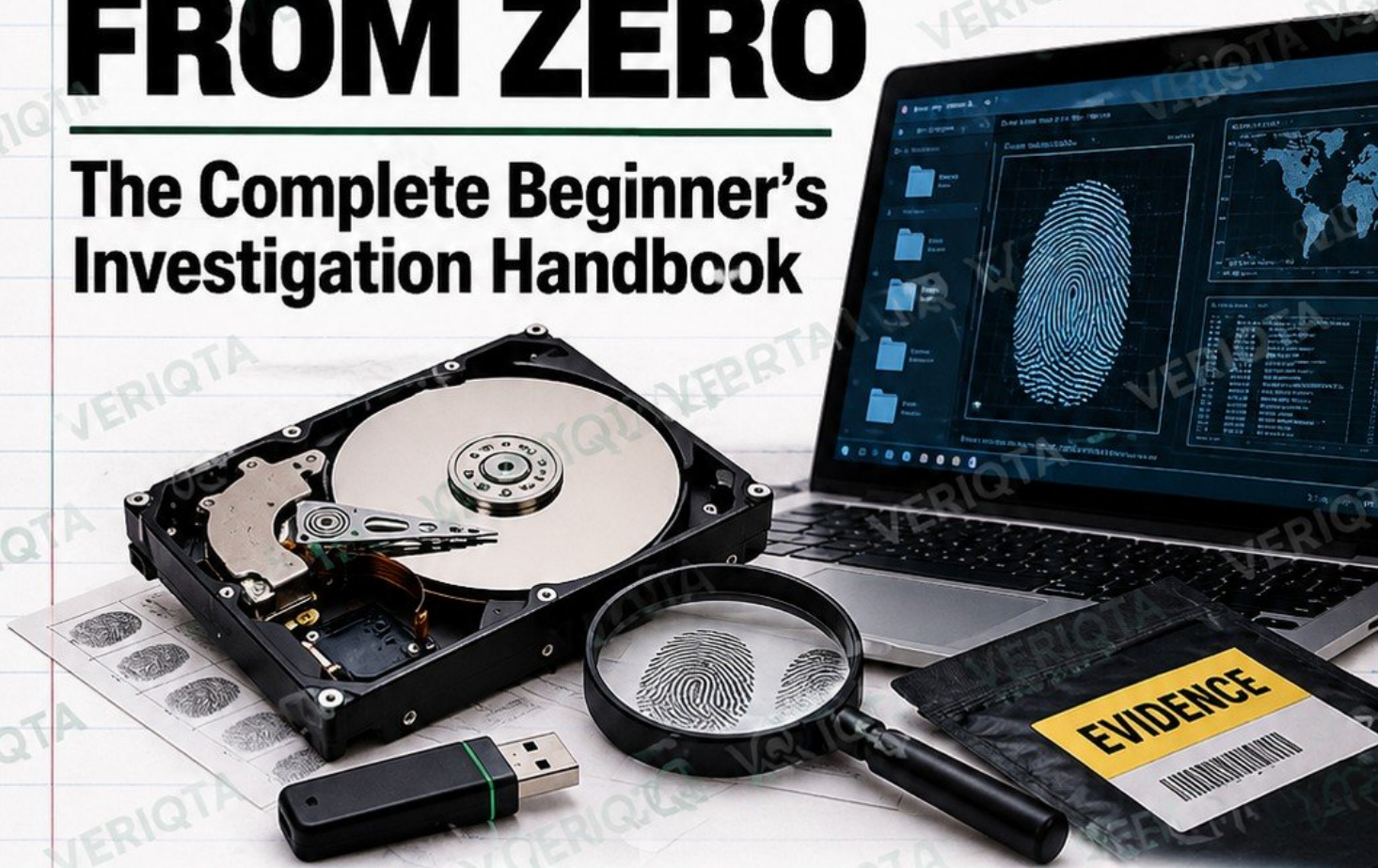


DIGITAL FORENSICS FROM ZERO

**The Complete Beginner's
Investigation Handbook**



**UNDERSTAND
DIGITAL
EVIDENCE**



**LEARN
INVESTIGATION
TECHNIQUES**



**EXPLORE
REAL-WORLD
TOOLS**



**ANALYSE
ARTIFACTS**



**BUILD
TIMELINES**



**CREATE
INVESTIGATION
REPORTS**

PRACTICAL | BEGINNER-FRIENDLY | REAL-WORLD EXAMPLES



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta

Welcome to Digital Forensics

What is Digital Forensics?

Digital forensics is the process of **finding, preserving, analysing and reporting digital evidence** from computers, mobile devices, networks and cloud systems. It helps investigators understand what happened, when it happened, and who was involved.

Why Does It Exist?

- Helps investigate cybercrimes and security incidents
- Provides evidence for legal processes
- Identifies the source and impact of an attack
- Supports internal investigations for policy violations
- Helps organisations prevent future incidents

Common Investigation Scenarios



Cyberattacks
(malware, ransomware, data theft)



Insider threats
(unauthorised access, data leakage)



Fraud investigations
(financial and digital fraud)



Mobile device investigations
(messages, calls, apps, location)



Cloud investigations
(logs, user activity, configuration)

How an Investigation Works



Digital Evidence Examples



Files and folders



Logs
(system, application, security logs)



Browser data
(history, downloads, cookies)



Mobile data
(messages, apps, location)



Emails
(headers, attachments)



Network records
(IP addresses, connections, traffic)

Junior Engineer Key Takeaways

- ✓ Digital forensics is about facts, not assumptions.
- ✓ Evidence must be handled carefully.
- ✓ Investigations follow a structured process.
- ✓ Forensics is used in real incidents every day.
- ✓ Good documentation is essential.



Remember This:

Digital forensics helps you answer three key questions:

What happened? When did it happen? Who was involved?



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta

Thinking Like a Digital Investigator

Evidence versus assumptions, investigative questions, and avoiding conclusions without proof.

What Does It Mean?

A digital investigator uses **facts, not assumptions**, to understand what happened. Every finding must be based on verifiable digital evidence, not personal opinions, guesses, or pressure from others.



Evidence vs Assumptions

Evidence

- ✓ Verifiable and factual
- ✓ Can be shown and tested
- ✓ Based on data from devices, logs, or systems
- ✓ Stands up in legal and professional reviews
- ✓ Answers what actually happened

Assumptions

- ✗ Based on personal opinions or guesses
- ✗ Not verifiable
- ✗ Can be wrong
- ✗ Leads to false conclusions
- ✗ Can damage investigations
- ✗ Should never be used as findings

Common Examples

Evidence

-  Log file showing successful login
-  Email header with source IP address
-  CCTV timestamp of access
-  Mobile device location records
-  File metadata with timestamps

Assumption

-  The user must have done it
-  It was probably a hacker
-  They were in the office at that time
-  The file was deleted intentionally
-  It's a malware infection

Key Investigative Questions

- 01 What happened?
- 02 When did it happen?
- 03 Where did it happen?
- 04 How did it happen?
- 05 Who was involved?
- 06 What systems or data are affected?
- 07 What evidence do we have?
- 08 What additional evidence is needed?

Avoiding Conclusions Without Proof

- ✗ Do not jump to conclusions.
- ✗ Do not rely on assumptions.
- ✗ Do not let pressure influence your findings.
- ✗ Do not ignore evidence that does not match your initial theory.
- ✗ Always validate with verifiable evidence.
- ✗ Keep an open and objective mindset.
- ✗ Be ready to update your findings as new evidence is discovered.

Junior Investigator Mindset

- ✓ Be curious and ask questions.
- ✓ Look for evidence, not opinions.
- ✓ Think critically and stay objective.
- ✓ Document what you see.
- ✓ Verify and validate your findings.
- ✓ Be patient and follow the evidence.
- ✓ Understand that the truth is built from data.
- ✓ Your job is to find facts, not to prove someone guilty.



@veriqta



@veriqta



@veriqta



@veriqta



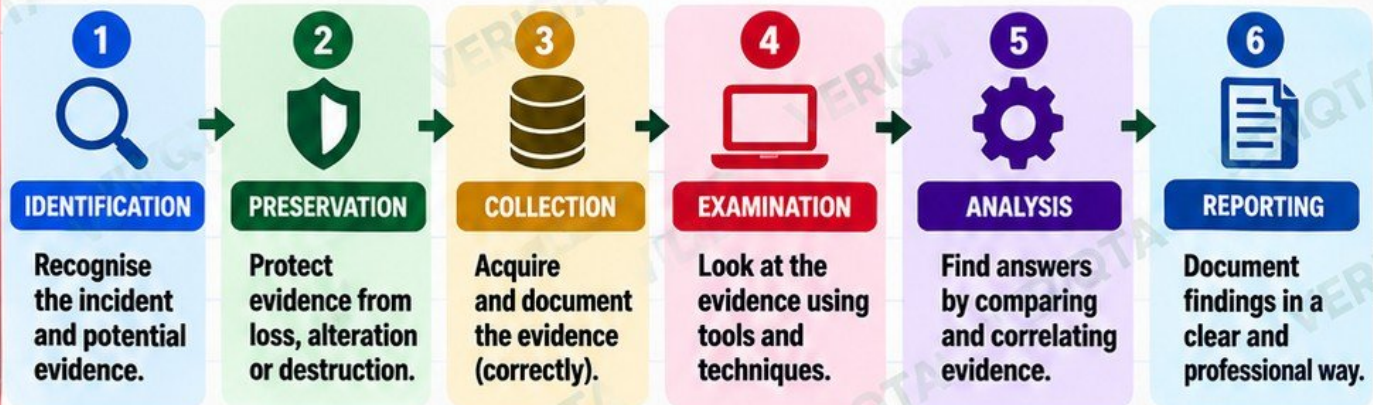
@veriqta



@veriqta

The Digital Investigation Lifecycle

Identification, preservation, collection, examination, analysis, and reporting



1. IDENTIFICATION



- Recognise a potential security incident or investigation need.
- Determine what type of evidence may exist.
- Define the scope and key questions.

Examples

- Suspicious login
- Malware infection
- Data breach
- Insider activity

2. PRESERVATION



- Protect evidence from being altered or lost.
- Isolate affected systems if needed.
- Use write blockers and forensic best practices.
- Document every action.

Key Points

- Do not power off without a plan.
- Prevent evidence destruction.
- Maintain chain of custody.

3. COLLECTION



- Acquire a forensic copy of the evidence.
- Collect from computers, mobile devices, networks, cloud, etc.
- Use approved tools and methods.
- Record details (date, time, tool, person).

Examples

- Disk image
- Memory dump
- Log files
- Network capture
- Mobile device extraction

4. EXAMINATION



- Explore the collected evidence using forensic tools.
- Recover files, view artifacts, check logs and metadata.
- Identify relevant data.
- Keep the original evidence unchanged.

Examples

- File system analysis
- Deleted file recovery
- Log review
- Browser artifacts
- Email analysis

5. ANALYSIS



- Interpret the findings to answer the investigation questions.
- Build timelines and correlate events.
- Identify the attacker's actions, methods and impact.
- Distinguish between facts and assumptions.

Key Questions

- What happened?
- When did it happen?
- How did it happen?
- Who was involved?
- What evidence proves it?

6. REPORTING



- Document the process, tools used, evidence examined and findings.
- Present a clear, factual and objective report.
- Include timelines, key evidence, conclusions and recommendations.
- Write for both technical and non-technical audiences.

Report Should Include

- Executive summary
- Scope and objectives
- Methodology
- Key findings
- Evidence details
- Limitations
- Recommendations



Important Notes

- ✓ Always follow legal and organisational policies.
- ✓ Maintain chain of custody.
- ✓ Use trusted and validated tools.
- ✓ Be objective and evidence-based.
- ✓ Do not make conclusions without proof.



Common Mistakes

- ✗ Changing or modifying original evidence.
- ✗ Relying on assumptions.
- ✗ Poor documentation.
- ✗ Using untrusted tools.
- ✗ Skipping steps in the lifecycle.

A SUCCESSFUL INVESTIGATION FOLLOWS A STRUCTURED LIFECYCLE FROM START TO FINISH.



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta

Understanding Digital Evidence

Files, logs, metadata, browser activity, memory, network records, and cloud artifacts

What is Digital Evidence?



Digital evidence is any data stored or transmitted in digital form that can be used to find out **what happened** during an incident or investigation.

Where Digital Evidence Can Be Found



Computers
(desktops,
laptops)



Mobile
devices



Servers
and cloud
systems



Networks
and logs



Cloud
services

Common Types of Digital Evidence



Files and Folders

Documents, images, executables, deleted files



Logs

System logs, application logs, security logs



Metadata

File details, timestamps, location, device info



Browser Activity

History, cookies, downloads, cached files



Memory (RAM)

Running processes, open files, credentials



Network Records

IP addresses, connections, DNS, traffic captures



Cloud Artifacts

User activity, logs, configurations, cloud storage

Examples and What They Can Reveal

Evidence Type

What It Can Reveal

Files and Folders

- Documents created or modified
- Deleted files and file history

Logs

- User logins and system activity
- Errors, security events, timelines

Metadata

- When a file was created or modified
- Device, author, and location info

Browser Activity

- Websites visited
- Downloads and searched terms

Memory (RAM)

- Running processes
- User activity and possible malware

Network Records

- IP addresses and connections
- Data sent or received

Cloud Artifacts

- User logins and file access
- Configuration and activity history

Why It Matters

- ✓ Helps investigators understand what happened.
- ✓ Provides a timeline of events.
- ✓ Identifies the source and impact of an attack.
- ✓ Supports legal and disciplinary actions.
- ✓ Helps prevent future incidents.



Common Beginner Mistakes

- ✗ Focusing only on deleted files.
- ✗ Ignoring logs and metadata.
- ✗ Assuming browser history tells the full story.
- ✗ Overlooking cloud and network evidence.
- ✗ Not preserving evidence before analysis.

Important Points for Beginners



- Digital evidence can exist in many places.
- Some evidence is visible, some is hidden.
- Evidence can be deleted but may still be recovered.
- Always consider timestamps and context.
- Different evidence types tell different parts of the story.

Junior Investigator Tip



Look at multiple sources of evidence. One piece of evidence rarely tells the whole story. Correlate files, logs, browser activity, and network records to understand the complete picture.



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta

The First 10 Minutes of an Incident

Initial observations, documenting system state, escalation, and avoiding evidence destruction.

Why the First 10 Minutes Matter?



- Critical evidence can be lost quickly.
- Your actions can preserve or destroy evidence.
- Early observations help identify the scope and severity.
- Good documentation makes the investigation easier.

What to Do Immediately

-  Observe and identify what is happening.
-  Document the current system state.
-  Preserve evidence and avoid changes.
-  Escalate to the right team.
-  Follow organisational incident process.

Initial Observations



- What system is affected?
- What are the symptoms?
- When did it start?
- What changed recently?
- Are other systems affected?
- Is there unusual network activity?

Document the System State



- Record the date and exact time.
- Note what you see on the screen.
- Take screenshots if possible.
- Record logged-in users.
- Note running processes and open connections.
- Capture system logs (read-only).
- Document any error messages.

Useful Commands (Read-Only)



- | | |
|-------------------------------|-----------------------------|
| <code>date</code> | Show current date and time |
| <code>who</code> | Show logged-in users |
| <code>ps aux</code> | List running processes |
| <code>netstat -tulnp</code> | Show network connections |
| <code>ss -tulnp</code> | Alternative to netstat |
| <code>dmesg tail</code> | View recent kernel messages |
| <code>journalctl -n 50</code> | View recent system logs |

Actions to Avoid



- Do not shut down or reboot the system.
- Do not delete files or logs.
- Do not run untrusted tools.
- Do not make configuration changes.
- Do not install software.
- Do not open or modify suspicious files.
- Do not connect the system to the internet.
- Do not assume what happened.

Escalation



- Report the incident to your security or IT team.
- Provide clear and factual information.
- Share the impact and affected systems.
- Follow your organisation's incident response plan.
- Work with the incident response or forensics team.

Junior Investigator Tips



- Stay calm and focused.
- Document everything, even small details.
- Take screenshots before doing anything.
- Use read-only commands where possible.
- Stick to facts, not assumptions.
- If in doubt, stop and ask your team.

Remember



Your goal in the first 10 minutes is to observe, document, preserve evidence, and escalate. Do not make changes. Let the investigation team perform deeper analysis.



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta

Chain of Custody Explained

Tracking who collected, accessed, transferred, or handled evidence.

What is Chain of Custody?

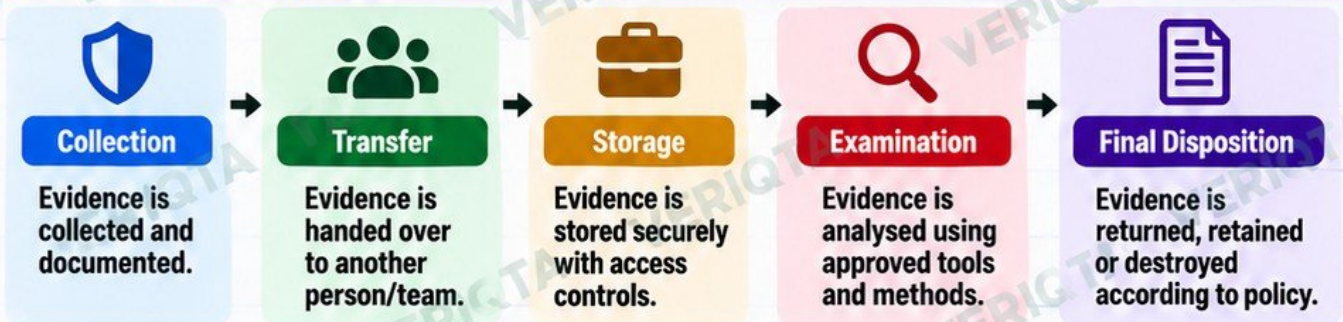


Chain of custody is the documented record of every person who collected, accessed, transferred, stored, examined, or handled evidence from the time it is collected until its final disposition.

Why It Is Important

- ✓ Proves the evidence is authentic.
- ✓ Shows the evidence has not been altered or tampered with.
- ✓ Establishes accountability.
- ✓ Required for legal and professional investigations.

Chain of Custody Flow



Chain of Custody Record Information

Field	Details to Record
 Date and time	When the action occurred
 Person's name	Who handled the evidence
 Organisation	Their organisation or department
 Action	Collected, transferred, accessed, stored, examined, returned, destroyed
 Evidence description	Type, identifier, and details
 Location	Where the evidence was
 Purpose	Why the evidence was handled
 Signature	Person's signature or digital approval

Example Chain of Custody Log

Date/Time	Name	Action	Location	Notes
2026-10-01 09:15	A. Felix	Collected	Office PC	Seized laptop SN: ABC123
2026-10-01 10:00	J. Smith	Transferred	Forensics Lab	Handed over for imaging
2026-10-01 10:20	K. Brown	Accessed	Forensics Lab	Created forensic image
2026-10-01 12:30	K. Brown	Examined	Forensics Lab	Analysed disk image (read-only)
2026-10-02 14:00	A. Felix	Returned	Evidence Storage	Stored in secure locker E-01

Common Mistakes

- ✗ Not documenting every handover.
- ✗ Missing dates, times or details.
- ✗ Allowing unauthorised access.
- ✗ Changing or modifying the original evidence.
- ✗ Incomplete or unclear records.
- ✗ Storing evidence in an insecure location.

Best Practices

- ✓ Document every action immediately.
- ✓ Use clear, accurate and consistent information.
- ✓ Keep evidence in secure, access-controlled storage.
- ✓ Use tamper-evident seals where applicable.
- ✓ Follow organisational policies and legal requirements.
- ✓ Maintain a complete and unbroken chain of custody.



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta

Evidence Integrity and Hashing

Understanding SHA-256, verifying evidence integrity, and using sha256sum.

What is a Hash?



- A hash is a unique fixed-length string generated from a file using a mathematical algorithm.
- Even a tiny change in the file will produce a completely different hash value.
- Hashes help verify that evidence has not been altered.

Why Hashing is Important

- ✓ Proves the integrity of evidence.
- ✓ Detects if a file has been modified.
- ✓ Used in chain of custody records.
- ✓ Accepted in legal and professional investigations.
- ✓ Helps ensure the evidence you analyse is exactly the same as the original.

What is SHA-256?



SHA-256

- SHA-256 is a widely used cryptographic hash algorithm.
- It produces a 256-bit (64 character) hash value.
- It is commonly used in digital forensics to verify file integrity.

How Hashing Works



Input File
(any size)

SHA-256
Algorithm



a3f5e1...
(64 characters)
Hash Value

Generate a SHA-256 Hash

```
# Calculate the SHA-256 hash of a file
sha256sum evidence.img
```

Example Output:

```
a3f5e1d7c9b8a6f4e2d1c7b9e8f5a6d4c3e2b1a9f8e7d
6c5b4a3210f9e8d7c6b5a4f3210e9d8c7 evidence.img
```

Verify a Hash

```
# Verify the hash of a file
sha256sum -c hashfile.txt
```

Example hashfile.txt content:

```
a3f5e1d7c9b8a6f4e2d1c7b9e8f5a6d4c3e2b1a9f8e7d
6c5b4a3210f9e8d7c6b5a4f3210e9d8c7 evidence.img
```

Example Output (if file is unchanged):

```
evidence.img: OK
```

Example Output (if file is modified):

```
evidence.img: FAILED
```

Hashing Different File Types

File Type	Command Example	Use Case
 Disk image	sha256sum disk.img	Verify forensic image
 Log file	sha256sum auth.log	Check log integrity
 Document	sha256sum report.pdf	Verify report file
 Executable	sha256sum malware.exe	Confirm file has not changed
 Any file	sha256sum filename	General use

Common Mistakes



- Using the wrong algorithm (e.g., MD5 instead of SHA-256).
- Calculating the hash after the file has been modified.
- Not recording the hash value in the chain of custody.
- Assuming a file is unchanged without verification.

Important Notes



- Always calculate the hash immediately after collection.
- Record the hash value in your chain of custody.
- Use trusted tools and verified algorithms.
- Compare hashes before and after analysis.
- A different hash means the file has been modified.
- Store hash values securely.

Best Practices



- Always use SHA-256 for investigations.
- Verify hashes at multiple stages.
- Keep a record of hash values with date, time and person.
- Document the tools and commands used.
- Store original evidence and hash values securely.
- Include hash values in your final report.



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta

Disk Imaging Fundamentals

Original media versus forensic copies, write blockers, and acquisition methods.

What is Disk Imaging?



- Disk imaging is the process of creating an exact, bit-for-bit copy of a storage device.
- The copy is called a forensic image.
- It captures all data, including deleted files, unallocated space, and system artifacts.
- The original media must be preserved and not altered.

Original Media vs Forensic Copy

Original Media (Source)



- Actual physical device (HDD, SSD, USB, etc.).
- Must be protected from changes.
- Not used for analysis (if possible).

Forensic Copy (Image)



- Exact bit-for-bit copy of the original.
- Used for analysis.
- Does not alter the original.
- Can be verified using hash values.

Why Use a Write Blocker?



- Prevents any write operations to the original media.
- Ensures the evidence remains unchanged.
- Used for both HDD and SSD.
- Hardware write blockers are preferred for critical evidence.

Disk Imaging Process



1. Connect with write blocker

2. Acquire forensic image

3. Verify hash values

4. Analyse the copy (not original)

Common Acquisition Methods

Method	Description	Use Case
Bit-for-Bit Image	Exact copy of all sectors (including unallocated space).	Standard method for most investigations.
Logical Image	Copies only active files and file system data.	Useful for quick acquisition (less complete).
Live Imaging	Image a running system without shutting it down.	Used when system must remain online.
Physical Image	Complete sector-by-sector copy of the entire device.	Preferred for full evidence preservation.

Common Tools for Disk Imaging

Tool	Platform	Description
FTK Imager	Windows	Widely used forensic imaging tool.
dcfldd	Linux	Forensic version of dd with hashing.
Guymager	Linux	User-friendly forensic imaging tool.
dd	Linux	Command-line disk imaging tool.
EnCase	Windows	Commercial forensic imaging and analysis.

Common Mistakes



- Imaging without a write blocker.
- Not verifying hash values.
- Analysing the original media.
- Using incorrect acquisition settings.
- Poor documentation.

Example: Creating a Disk Image (Linux)

```
# Create a forensic image using dd
sudo dd if=/dev/sdb of=/evidence/disk.img \
bs=4M status=progress
```

Verify the image hash:

```
sha256sum /evidence/disk.img
```

Example output:

```
a3f5e1d7c9b8a6f4e2d1c7b9e8f5a6d4c3e2b1a9f8e7d6
c5b4a3210f9e8d7c6b5a4f3210e9d8c7 disk.img
```

Important Notes



- Always use a write blocker when acquiring images.
- Calculate and record hash values (e.g., SHA-256).
- Store the original media in a secure location.
- Work on the forensic copy, never the original.
- Document the imaging process, tools, settings, and hash values.

Best Practices



- Use trusted and validated tools.
- Record tool version and configuration.
- Verify hashes immediately after acquisition.
- Keep multiple copies of the forensic image.
- Maintain a detailed chain of custody record.



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta

Understanding File Systems

How NTFS, FAT32, ext4, files, directories, and metadata relate to investigations.

What is a File System?



- A file system is a method used by an operating system to organise, store, and manage files on a storage device.
- It controls how data is written, read, and retrieved.
- Different operating systems use different file systems (e.g., NTFS, FAT32, ext4).

Why File Systems Matter in Investigations



- File systems store information about files, directories, and metadata.
- They help you find, recover, and analyse data during investigations.
- Metadata can reveal when a file was created, modified, or accessed.
- Deleted files may still exist in the file system until overwritten.
- Understanding the file system structure helps in data recovery and timelines.

Common File Systems

NTFS



- Used by Windows.
- Supports large files and advanced features (e.g., permissions, encryption).
- Stores rich metadata (e.g., timestamps, file permissions).

FAT32



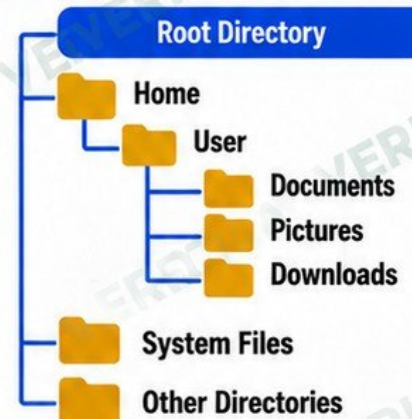
- Common on USB drives, external storage, and older systems.
- Simple structure.
- Supports smaller maximum file size (4 GB).
- Stores basic metadata.

ext4



- Common on Linux.
- Supports large files and high performance.
- Uses journaling for better data integrity.
- Stores metadata such as timestamps, ownership, and permissions.

File System Structure



Key Concepts

	File	A collection of data stored on disk.
	Directory	A folder used to organise files.
	Metadata	Information about a file (e.g., created, modified, accessed times, size, permissions).
	Allocation	How file data is stored on disk (clusters/blocks).
	Deleted File	A file marked as deleted but may still exist until overwritten.

Common Metadata in Investigations

Metadata Type	Description	Examples
Timestamps	When the file was created, modified, or accessed.	Created, Modified, Accessed (MAC times)
File Size	Size of the file in bytes.	e.g., 1,048,576 bytes
Permissions	Access rights for the file.	Read, Write, Execute
Owner	User or group that owns the file.	Username, Group name
File Type	Type or format of the file.	.docx, .jpg, .exe
Location	Path where the file is stored.	e.g., C:\Users\Documents
Other Attributes	Additional file information.	Hidden, System, Encrypted

Practical Investigation Uses



- Explore file system structure to find relevant files.
- Analyse metadata to build a timeline.
- Recover deleted files from unused space.
- Identify suspicious files based on location or type.
- Compare file timestamps to understand user activity.

Important Notes



- Different file systems store metadata differently.
- Deleted files may still be recoverable.
- Timestamps can be modified, so always corroborate with other evidence.
- Understanding file system behaviour helps prevent loss of important evidence.
- Use trusted forensic tools to analyse file systems.



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta

Recovering Deleted Evidence

How deletion works, recoverable artifacts, limitations, and why investigators avoid writing to source media.

How Deletion Works



- When a file is deleted, the operating system usually does not erase the data immediately.
- It marks the file's space as "available" for new data.
- The file may still exist on the disk until it is overwritten.
- Quick format removes file system references, but the data can often still be recovered.

Why Evidence Can Be Recovered



- File data often remains on disk until overwritten.
- File system metadata (e.g., MFT, inodes) still contains information.
- Traces exist in unallocated space.
- Artifacts may exist in logs, recent files lists, thumbnails, and backups.
- Even after deletion, remnants can be found in memory, shadow copies, or cloud sync data.

Common Recoverable Artifacts

Artifact Type	Description	Examples
 Deleted Files	File content still present in unallocated space.	Documents, images, videos, archives
 File System Metadata	Records about the file (even if deleted).	MFT (NTFS), inodes (ext4), directory entries
 Timestamps	Created, modified, accessed, deleted times.	MAC times, \$LogFile, \$UsnJrnl (NTFS)
 Application Artifacts	Traces in application histories and caches.	Browser history, recent files, thumbnails
 System Logs	Logs may show file activity or deletion.	Windows Event Logs, Linux logs, macOS logs
 Backups and Copies	Previous versions or sync copies.	Shadow copies, cloud sync (OneDrive, Google Drive), backups

Why Investigators Avoid Writing to Source Media



- Writing new data can overwrite deleted files and make recovery impossible.
- It can alter timestamps and metadata (e.g., MAC times).
- It may destroy valuable artifacts in unallocated space.
- Investigators use write blockers and work on forensic copies instead of the original media.

Common Recovery Tools (Examples)

Tool	Platform	Use Case
Autopsy	Windows/Linux/macOS	Recover deleted files and analyse artifacts
PhotoRec	Windows/Linux/macOS	Recover deleted files by file signature
TestDisk	Windows/Linux/macOS	Recover lost partitions and files
Recuva	Windows	Basic file recovery for Windows
Extundelete	Linux	Recover deleted files from ext2/ext3/ext4

Limitations of Data Recovery



- If the data has been overwritten, it may be unrecoverable.
- Not all deleted files can be restored.
- Recovery may be partial or incomplete.
- Some file types are easier to recover than others.
- Solid-state drives (SSDs) with TRIM can make recovery much harder.
- Encryption may prevent access to recovered data without the decryption key.

Key Takeaways



- Deletion does not always mean the data is gone.
- Many artifacts can still be recovered.
- Always work on a forensic copy, never the original.
- Use write blockers to protect the source media.
- Understand the limitations of recovery.
- Combine multiple sources of evidence for a complete analysis.

Best Practices



- Use a write blocker when accessing the media.
- Create a forensic image before analysis.
- Use verified forensic tools.
- Document every step of the recovery process.
- Preserve metadata and timestamps.
- Be aware of file system limitations and SSD behaviour.
- Correlate recovered data with other artifacts.



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta

Investigating Windows Artifacts

Event Logs, Registry, Prefetch, Recycle Bin, and user activity.

What are Windows Artifacts?



- Windows artifacts are traces of user activity, system activity, and application usage.
- They are created by the operating system, applications, and users.
- These artifacts help investigators reconstruct what happened on a Windows system.

Why They Are Important



- Provide evidence of user actions and timelines.
- Help identify malicious activity.
- Can show how a system was used.
- Useful for incident response, forensic investigations, and legal cases.

Key Windows Artifacts

Artifact	Location	What It Contains	Example Use Cases
 Event Logs	C:\Windows\System32\winevt\Logs\	System, security and application events.	Logon activity, system changes, error events, suspicious activity.
 Registry	C:\Windows\System32\config\	Configuration settings, user activity, recently used files, installed software.	User activity, USB devices, program execution, system settings.
 Prefetch	C:\Windows\Prefetch\	Records of executed applications (with timestamps).	Applications run, time of execution, user activity timeline.
 Recycle Bin	C:\\$Recycle.Bin\	Deleted files and original file information.	Recover deleted files, file names, original location and timestamps.
 User Activity	C:\Users\<username>\	Recent files, documents, browser activity, application logs.	User behaviour, recently opened files, browser history, document activity.

Windows Event Logs



- Located in Event Viewer (eventvwr.msc).
- Types: Security, System, Application.
- Contain important events such as logon/logoff, account changes, system errors, and more.
- Event IDs help identify specific activities (e.g., 4624 for successful logon).

Windows Registry



- Stores configuration and user activity data.
- Useful locations:
 - Recent files: NTUSER.DAT
 - USB devices: SYSTEM hive
 - Installed programs: SOFTWARE hive
 - User activity: HKCU\Software\Microsoft\Windows\CurrentVersion\Explorer
- Can reveal executed programs, MRU lists, and system configuration changes.

Prefetch Files



- Located in C:\Windows\Prefetch\
- Record information about executed applications.
- Contain the application name, last run time, and number of executions.
- Useful for building a timeline of application usage.

Recycle Bin



- Deleted files are moved to the Recycle Bin before permanent deletion.
- Metadata such as original path, deletion time, and file size can be recovered.
- Files can often be restored if not overwritten.

User Activity Artifacts



- Recent files: C:\Users\<user>\AppData\Roaming\Microsoft\Windows\Recent\
- Browser history: Edge, Chrome, Firefox
- Jump Lists, LNK files, search history
- User documents, pictures, and downloads
- Provide a clear timeline of user behaviour.

Best Practices



- Work on a forensic copy, not the original system.
- Use trusted forensic tools.
- Correlate multiple artifacts for a complete timeline.
- Validate timestamps and time zones.
- Document every step and maintain chain of custody.
- Be aware of anti-forensics techniques.



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta

Investigating Linux Artifacts

Authentication logs, journalctl, shell history, processes, and suspicious activity.

What are Linux Artifacts?



- Linux artifacts are traces of user activity, system activity, and application usage.
- They are stored in log files, configuration files, and system databases.
- These artifacts help investigators understand what happened on a Linux system.

Why They Are Important



- Show evidence of user logins, commands, and actions.
- Help identify malicious activity, unauthorised access, and persistence.
- Provide a timeline of events.
- Useful for incident response, forensic investigations, and compliance.

Key Linux Artifacts

Artifact	Location (Common Paths)	What It Contains	Example Use Cases
 Authentication Logs	/var/log/auth.log /var/log/secure	Login attempts, SSH activity, sudo usage, authentication failures.	Identify brute force attacks, unauthorised logins, and user activity.
 System Logs (journalctl)	journalctl /var/log/*	System events, service logs, boot logs, kernel messages.	Investigate system activity, service starts/stops, and errors.
 Shell History	~/.bash_history ~/.zsh_history	Commands executed by users.	Find suspicious commands, malicious activity, and lateral movement.
 Processes	ps, top, /proc/	Running processes, parent/child relationships, command line arguments.	Identify malicious or unusual processes.
 Cron Jobs	/etc/crontab /var/spool/cron/	Scheduled tasks and scripts.	Detect persistence and suspicious scheduled jobs.
 User and System Configuration	/etc/passwd /etc/shadow /etc/sudoers	User accounts, password info, privileges and configurations.	Find new users, privilege escalation, and changes to configuration.
 Network Artifacts	/var/log/syslog /var/log/messages ss, netstat, /proc/net/	Network connections, listening ports, firewall logs.	Identify suspicious connections, exfiltration, or C2 traffic.

Using journalctl

```
# View all logs
journalctl

# View logs for a specific service
journalctl -u ssh

# View logs within a time range
journalctl --since "2024-01-01" --until "2024-01-02"
```

Checking Authentication Logs



- View failed login attempts
sudo cat /var/log/auth.log | grep "Failed"
- View successful logins
sudo cat /var/log/auth.log | grep "Accepted"
- Check sudo usage
sudo cat /var/log/auth.log | grep "sudo"

Analysing Shell History



```
# View user shell history
cat ~/.bash_history

# Look for suspicious commands
grep -i "wget|curl|nc|base64" \
~/.bash_history

# Check history timestamps (if enabled)
cat ~/.bash_history | nl
```

Investigating Running Processes



- # View running processes
ps aux
- # Real-time process monitoring
top
- # Check process details
ps -fp <PID>
- # Look for suspicious processes
grep -i "malwarename" /proc/*/cmdline

Detecting Suspicious Activity



- Unusual login times or locations.
- Multiple failed login attempts.
- Unknown or unexpected processes.
- Suspicious network connections.
- Unexpected cron jobs.
- Changes to critical system files.
- Use of known attack tools (e.g., nc, nmap, msf, curl).

Best Practices



- Always work on a forensic copy (not the original system).
- Use trusted forensic tools.
- Correlate multiple artifacts.
- Look for unusual patterns or behaviour.
- Validate timestamps (consider timezone).
- Document all findings clearly.
- Maintain chain of custody.
- Be aware of anti-forensics techniques.



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta

Memory Forensics for Beginners

What RAM can reveal, volatile evidence, memory acquisition, and an introduction to Volatility 3.

What is Memory Forensics?



- Memory forensics is the analysis of data from a computer's RAM (Random Access Memory).
- It helps investigators find evidence that may not exist on disk.
- It captures the current state of a system, including running processes, network connections, loaded modules, and user activity.

Why RAM is Important



- Contains volatile (temporary) evidence.
- Shows what was running at the time of acquisition.
- Can reveal malicious activity, hidden processes, and injected code.
- Helps answer questions that disk logs cannot, such as active network connections, encryption keys, and in-memory malware.

What RAM Can Reveal

	Running Processes	List of active processes, including suspicious or hidden processes.
	Network Connections	Active network connections, IP addresses, and open ports.
	Credentials	Plain text passwords, tokens, and encryption keys (sometimes).
	Loaded Modules	DLLs, drivers, and injected libraries.
	User Activity	Currently logged-in users and recent commands.
	Malware Indicators	In-memory malware, rootkits, and process injection.

Volatile vs Non-Volatile Evidence

Volatile Evidence (In Memory)



- Processes
- Network connections
- Logged-in users
- Open files
- Encryption keys
- Malware in memory
- System state

Lost when the system is powered off.

Non-Volatile Evidence (On Disk)



- Log files
- Registry files
- Configuration files
- User data
- Installed programs
- Timestamps
- Deleted files (sometimes recoverable)

Remains after reboot (unless deleted).

Memory Acquisition Methods



- Use trusted and verified tools.
- Common tools: DumpIt (Windows), LiME (Linux), FTK Imager, WinPMEM.
- Save the memory image to external storage.
- Calculate and record hash values (e.g., SHA-256).
- Follow proper chain of custody procedures.

Introduction to Volatility 3



- Volatility 3 is a powerful open-source memory forensics framework.
- It supports multiple operating systems (Windows, Linux, macOS).
- It helps analyse memory images to extract processes, network connections, modules, files, and more.
- It uses plugins to perform different analysis tasks.

Common Volatility 3 Commands



```
# Check image information
vol.py -f memory.img windows.info
# List running processes
vol.py -f memory.img windows.pslist
# Show network connections
vol.py -f memory.img windows.netscan
# Extract command line arguments
vol.py -f memory.img windows.cmdline
# List loaded modules
vol.py -f memory.img windows.modules
```

Key Takeaways



- RAM can reveal critical evidence not found on disk.
- Volatile evidence is lost when the system is shut down.
- Use trusted tools to acquire memory images.
- Always verify hash values and document the process.
- Volatility 3 is a key tool for analysing memory images.

Best Practices



- Acquire memory as early as possible.
- Use trusted and validated tools.
- Work on a copy of the memory image.
- Document every step and calculate hash values.
- Combine memory analysis with disk and log analysis.
- Follow legal and organisational guidelines.



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta

Network Forensics Fundamentals

IP addresses, DNS, connections, packet captures, and Wireshark.

What is Network Forensics?



- Network forensics is the capture, analysis, and investigation of network traffic.
- It helps identify suspicious activity, attacks, data exfiltration, and policy violations.
- It provides evidence of what happened on a network, who was involved, and when it occurred.

Why Network Forensics is Important



- Helps detect and investigate cyber attacks (e.g., malware, C2, data theft).
- Provides visibility into network activity and user behaviour.
- Can identify unusual connections, suspicious DNS queries, and unauthorised access.
- Supports incident response, threat hunting, and legal investigations.

Key Networking Concepts

Concept	Description
 IP Address	Unique identifier for a device on a network (IPv4 or IPv6).
 DNS	Translates domain names (e.g., google.com) to IP addresses.
 Ports	Identify the service or application (e.g., 80 = HTTP, 443 = HTTPS).
 Protocols	Rules for communication (e.g., TCP, UDP, HTTP, DNS, ICMP).
 Connections	Source IP, destination IP, source port, destination port, protocol and session state.

Common Network Artifacts

Artifact	What It Shows
 Packet Captures (PCAP)	Full network traffic (requests, responses, files, communications).
 DNS Logs	Domain queries and resolved IP addresses.
 Firewall Logs	Allowed/blocked connections, source/destination IPs, ports.
 Proxy Logs	Web access, domains visited, user activity.
 IDS/IPS Logs	Detected threats and suspicious network activity.
 Router/Switch Logs	Network events, connection changes, errors.

Packet Capture with Wireshark



- Wireshark is a popular tool for capturing and analysing network traffic.
- You can capture live traffic or open a saved PCAP file.
- It allows you to filter, search, and inspect network packets.
- Useful for identifying malicious activity, data transfers, and protocol behaviour.

Common Wireshark Filters

Filter	Purpose
ip.addr == 192.168.1.10	Filter traffic to/from a specific IP address.
dns	Show only DNS traffic.
tcp.port == 80	Show HTTP traffic.
http	Show HTTP requests and responses.
tcp.flags.syn == 1	Show TCP SYN packets (connection attempts).
ip.addr == 10.0.0.5 && http	Filter HTTP traffic for a specific IP.

Investigating Network Activity



- Identify suspicious IP addresses.
- Analyse DNS queries (unusual or malicious domains).
- Look for unusual connections (external IPs, uncommon ports).
- Inspect HTTP/HTTPS traffic (requested domains, files).
- Look for indicators of data exfiltration.
- Correlate with firewall, proxy, and IDS/IPS logs.

Best Practices



- Capture traffic as early as possible during an incident.
- Use proper filters to focus on relevant traffic.
- Preserve original PCAP files and maintain chain of custody.
- Correlate network data with host and log data.
- Be aware of encrypted traffic (HTTPS, VPN) and its limitations.
- Use trusted tools and keep them updated.



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta

Browser and Email Forensics

Browser history, downloads, timestamps, email headers, and phishing evidence.

What is Browser and Email Forensics?



- Browser forensics involves analysing web activity stored on a device.
- Email forensics involves examining email content, headers, and related artifacts.
- These artifacts can provide evidence of user activity, communication, file transfers, and potential malicious activity.

Why It Is Important



- Helps reconstruct user activity and timelines.
- Can reveal evidence of data theft, phishing, malware, and unauthorised access.
- Provides valuable evidence for incident response, investigations, and legal cases.
- Helps identify who was involved, what was accessed, and when it happened.

Common Browser Artifacts

Artifact	What It Contains	Location (Examples)
 History	Visited websites and timestamps	Chrome, Edge, Firefox: History/places databases
 Downloads	Downloaded files, URLs, timestamps	Downloads database, browser history files
 Cookies	User sessions and authentication tokens	Cookies database (e.g., Cookies, places.sqlite)
 Cache	Cached web content (images, files, scripts)	Cache directories, browser cache files
 Form Data	Search queries, entered text, usernames	Form history database
 Saved Passwords	Stored usernames and passwords (encrypted)	Login Data database (encrypted, can be decrypted with tools)

Browser Forensic Tools (Examples)

Tool	Platform	Use Case
Chrome/Edge/Firefox Built-in Tools	Windows/Linux/macOS	View history, downloads, cookies, cache.
Browser History Viewer	Windows	Analyse browser databases.
BrowsingHistoryView	Windows	Quick view of browsing history.
HstEx	Windows	Advanced browser history analysis.
Autopsy	Windows/Linux/macOS	Analyse browser artifacts as part of a full investigation.

Key Takeaways



- Browser and email artifacts can reveal user activity, communication, and malicious behaviour.
- Timestamps help build accurate timelines.
- Email headers show the true source and path of a message.
- Be alert for phishing indicators such as spoofed domains and malicious attachments.
- Use trusted forensic tools and always maintain chain of custody.

Common Email Artifacts

Artifact	What It Contains	Example Sources
 Email Headers	Sender, recipient, subject, routing details, timestamps	Outlook (.msg), Thunderbird (.eml), Gmail (headers)
 Email Content	Full message body, attachments	.msg, .eml, PST files
 Attachments	Files sent or received (can contain malware)	Attachments folder, PST, .msg, .eml
 Trace Headers	Full email path (Received: fields)	View original headers in email client
 Phishing Indicators	Suspicious links, spoofed domains, malicious attachments	Email headers, URLs, sender details

Example Email Header Analysis

From: user@example.com
 To: victim@company.com
 Subject: Meeting Update
 Date: Mon, 10 Feb 2025 14:23:11 +0000
 Received: from mail.example.com (192.0.2.10) by mx.company.com with ESMTPS id ABC123 for <victim@company.com> Mon, 10 Feb 2025 14:23:11 +0000

→ Sender information
 → Date and time
 → Email path (shows how the email travelled)

Email Forensic Tools (Examples)

Tool	Platform	Use Case
Microsoft Outlook	Windows	View email headers and .msg files.
Mozilla Thunderbird	Windows/Linux/macOS	Analyse .eml files and headers.
MFCMAPI	Windows	Advanced analysis of Outlook PST files.
Email Header Analyzer (online tools)	Web	Analyse email headers and detect phishing indicators.
Autopsy	Windows/Linux/macOS	Analyse email artifacts in forensic images.

Best Practices



- Work on a forensic copy, not the original device.
- Use trusted and up-to-date forensic tools.
- Preserve timestamps and metadata.
- Validate findings with multiple artifacts.
- Look for signs of phishing, malware, and data exfiltration.
- Document every step and maintain chain of custody.



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta

Introduction to Forensic Tools

Autopsy, FTK Imager, Wireshark, Volatility 3, and appropriate use cases.

What are Forensic Tools?



- Forensic tools are specialised software used to collect, analyse, and preserve digital evidence.
- They help investigators examine disks, memory, network traffic, and system artifacts.
- Each tool has different strengths and is used for specific types of investigations.

Why They Are Important



- Help extract and analyse evidence in a structured and reliable way.
- Support incident response, forensic investigations, and legal cases.
- Save time and increase accuracy.
- Allow investigators to examine data without altering the original evidence (when used correctly).
- Each tool fills a specific role in the forensic process.

Overview of Key Forensic Tools

Tool	Purpose	Key Features	Use Cases	Platform
 Autopsy	Digital forensics platform for analysing disk images.	• File system analysis • Timeline analysis • Keyword search • File carving	• Analyse disk images • Investigate file activity • Build timelines • Identify deleted files	Windows Linux macOS
 FTK Imager	Create forensic disk images and verify integrity.	• Disk and partition imaging • Hash value verification • Preview files in images • Supports multiple formats	• Acquire disk images • Verify evidence integrity • Preview evidence • Work with raw or E01 images	Windows
 Wireshark	Network protocol analysis and packet capture.	• Capture and analyse packets • Inspect protocols (HTTP, DNS, etc.) • Follow streams • Filter and search traffic	• Analyse network traffic • Investigate malicious connections • Identify data exfiltration • Analyse suspicious communication	Windows Linux macOS
 Volatility 3	Memory forensics framework for analysing RAM images.	• Analyse memory dumps • Detect running processes • Find malicious artifacts • Supports multiple operating systems	• Investigate in-memory malware • Analyse running processes • Find hidden activity • Extract memory artifacts	Windows Linux macOS

Common Use Cases

Autopsy	FTK Imager	Wireshark	Volatility 3
			
• File system analysis • Timeline creation • Recover deleted files • Investigate user activity	• Acquire disk images • Verify hash values • Preview evidence • Create E01 or raw images	• Capture network traffic • Analyse protocols • Investigate suspicious communication • Identify data exfiltration	• Analyse memory dumps • Detect malware • Find hidden processes • Extract in-memory artifacts

Key Commands / Examples

Tool	Example Command / Action
FTK Imager	Create a disk image and verify hash (MD5/SHA1).
Wireshark	Capture packets and apply display filters (e.g., http, dns, ip.addr == 192.168.1.10).
Volatility 3	vol.py -f memory.img windows.pslist vol.py -f memory.img windows.netscan
Autopsy	Load a disk image, run ingest modules, search for keywords, and generate timeline.

Best Practices



- Use trusted and verified versions of forensic tools.
- Work on forensic copies, not the original evidence.
- Verify hash values to maintain integrity.
- Use the right tool for the right type of evidence.
- Document every step and maintain chain of custody.

Key Takeaways



- Autopsy is for disk analysis and timeline investigation.
- FTK Imager is for acquiring and verifying disk images.
- Wireshark is for capturing and analysing network traffic.
- Volatility 3 is for memory forensics and in-memory analysis.
- Each tool has specific use cases and complements the others.



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta

Building an Investigation Timeline

Connecting timestamps, events, logs, and artifacts to reconstruct activity.

What is an Investigation Timeline?



- An investigation timeline is a chronological record of events based on evidence from multiple sources.
- It connects timestamps, logs, artifacts, and user activity to show what happened, when it happened, and how events are related.
- It helps investigators understand the full sequence of activity and identify key actions, suspicious behaviour, and potential attackers.

Why It Is Important



- Reconstructs the sequence of events.
- Combines evidence from different sources (system logs, browser data, network logs, etc.).
- Helps identify malicious activity, data theft, and unauthorised access.
- Supports incident response, reporting, and legal cases.
- Provides a clear and defensible story of what happened.

Common Sources of Timestamps

Source	Examples	What It Provides
 Windows Logs	Event Logs (Sysmon, Security, System)	Logon events, process creation, system changes
 Linux Logs	auth.log, syslog, journalctl	Logins, commands, service activity
 Browser Data	History, downloads, cookies, cache	Web activity, file downloads, sites visited
 File System Metadata	MAC times (Modified, Accessed, Created)	File and folder activity timestamps
 Network Logs	Firewall logs, proxy logs, PCAP (Wireshark)	Connections, IP addresses, protocols, data transfers
 Email Artifacts	Email headers, sent/received times, attachments	Communication activity, sender/recipient, timelines
 Application Logs	RDP, VPN, Office, Cloud logs	User activity, document access, remote connections

Example Investigation Timeline

Date & Time	Event	Source	Details
2025-02-10 08:15:23	User login	Windows Event Log	Successful login for user "jdoe"
2025-02-10 08:16:05	Process execution	Sysmon	cmd.exe launched
2025-02-10 08:18:10	File download	Browser History	malicious_file.exe downloaded
2025-02-10 08:19:45	Network connection	Firewall Log	Connection to 185.199.110.10 (port 443)
2025-02-10 08:20:12	Program execution	File System Metadata	malicious_file.exe executed
2025-02-10 08:25:30	Email sent	Email Log	Suspicious email with attachment sent to external user
2025-02-10 08:28:10	Persistence	Registry Log	Run key added for malicious file

Steps to Build a Timeline

- 1 Identify relevant data sources (logs, artifacts, network captures).
- 2 Extract and normalise timestamps (convert to the same time zone).
- 3 Collect and correlate events from different sources.
- 4 Sort events in chronological order.
- 5 Identify key activities (e.g., initial access, execution, data transfer).
- 6 Validate the timeline and fill in any gaps.
- 7 Present a clear and concise timeline with evidence references.

Key Takeaways



- A timeline connects timestamps, events, and artifacts.
- Use multiple sources to get a complete picture.
- Always normalise timestamps to the same time zone.
- Look for patterns and suspicious activity.
- A clear timeline strengthens investigation findings and reports.

Tools for Timeline Analysis

Tool	Purpose	Use Case
Plaso	Create a super timeline from multiple sources	Analyse large amounts of forensic data
Timesketch	Visualise and explore timelines	Investigate and share timelines
Autopsy	Timeline analysis and artifact correlation	Analyse disk images
Log2Timeline	Convert logs to timeline format	Process log files for timeline analysis
Spreadsheet (Excel)	Manual timeline creation and analysis	Small investigations or custom analysis

Best Practices



- Use trusted and verified time sources (e.g., UTC).
- Correlate events from multiple data sources.
- Document the source and evidence for each event.
- Be aware of timestamp manipulation by attackers.
- Validate the timeline and review for inconsistencies.
- Keep a clear chain of custody for all evidence.



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta

Investigating a Suspicious Login

A guided fictional case involving unusual authentication activity and supporting evidence.

Case Scenario



- A user reports that they may have been hacked after seeing an unfamiliar login.
- Security alerts show a login from a new country and device at an unusual time.
- Your task is to investigate the suspicious login, identify what happened, and gather supporting evidence.

Investigation Objectives



- Confirm whether the login was legitimate or malicious.
- Identify the source (IP address, location, device, and time).
- Find supporting evidence from logs, browser data, and email.
- Determine the attacker's actions (if any).
- Document your findings in a clear timeline.

Investigation Steps

- 1 Review Authentication Logs**
Check Windows Event Logs, VPN logs, or cloud identity logs for login details.
- 2 Analyse IP Address and Location**
Identify the IP, location, and ISP. Check if it is known or suspicious.
- 3 Investigate the Device and Browser**
Look for device information, browser type, and user agent.
- 4 Check for Additional Activity**
Look for file access, downloads, or further suspicious actions after the login.
- 5 Correlate Evidence**
Combine logs, browser history, email alerts, and network data.
- 6 Build a Timeline**
Arrange events in chronological order to understand the full picture.

Sample Authentication Log

Time (UTC)	Event	IP Address	Location
2025-02-10 08:14:22	Successful login	41.203.15.10	Lagos, Nigeria
2025-02-10 19:42:10	Failed login (attempt)	185.199.110.20	Moscow, Russia
2025-02-10 19:45:03	Successful login	185.199.110.20	Moscow, Russia
2025-02-10 19:46:31	Password change	185.199.110.20	Moscow, Russia
2025-02-10 19:47:12	Email settings modified	185.199.110.20	Moscow, Russia

Supporting Evidence

-  **Authentication logs** (Windows, VPN, or cloud)
-  **IP address lookup** (location, ISP, reputation)
-  **Browser data** (history, cookies, device info)
-  **Email alerts** (login notifications, password change)
-  **File system evidence** (recent files, downloads)
-  **Network logs** (connections, firewall logs)

Investigation Timeline (Example)

08:14:22	User logged in from known location (Lagos, Nigeria).
19:42:10	Failed login attempt from Moscow, Russia (wrong password).
19:45:03	Successful login from Moscow, Russia.
19:46:31	Password changed by attacker.
19:47:12	Email settings modified (rule added to forward emails).
19:50:10	Suspicious email sent to external address.
20:05:33	User account used to access cloud files.
20:18:45	Attacker logged out.

Key Findings (Example)



- Login from an unfamiliar IP address in Russia.
- Multiple failed login attempts before success.
- Password changed after successful login.
- Email settings modified to forward emails.
- Suspicious files downloaded from cloud storage.
- Activity pattern indicates account compromise.

Key Takeaways



- Always investigate unusual logins, even if the user is unsure.
- Correlate evidence from multiple sources.
- Build a clear timeline to understand the full incident.
- Look for attacker actions after the initial login.
- Document findings with evidence.
- Use the timeline to support incident response and reporting.

Best Practices



- Monitor for unusual login activity.
- Use multi-factor authentication (MFA).
- Review logs regularly.
- Investigate alerts immediately.
- Preserve evidence and maintain chain of custody.
- Combine logs, network data, and endpoint evidence.



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta

Writing a Forensic Investigation Report

Scope, methodology, evidence inventory, findings, limitations, and recommendations.

1 What is a Forensic Report?



- A formal written document that presents the findings of a digital forensic investigation.
- It explains what was investigated, how it was done, what evidence was found, and what it means.
- It is used for internal review, management decisions, or legal proceedings.

2 Key Objectives



- Clearly communicate the scope and purpose of the investigation.
- Document the methodology and tools used.
- Present a complete inventory of evidence.
- Explain findings in a clear and objective way.
- Highlight limitations and uncertainties.
- Provide actionable recommendations based on the findings.

3 Typical Report Structure

1	Executive Summary	High-level overview of the investigation and key findings.
2	Scope	What was investigated and what was out of scope.
3	Methodology	Steps taken, tools used, and standards followed.
4	Evidence Inventory	List of all collected evidence with details and hashes.
5	Findings	Detailed analysis and results.
6	Limitations	Constraints, missing data, or uncertainties.
7	Conclusions	Summary of what the evidence shows.
8	Recommendations	Suggested actions based on the findings.

4 Evidence Inventory Example

No.	Evidence Item	Source	Description	Hash (SHA256)
1	Disk Image	Laptop (Windows 11)	Full disk image (500 GB)	a3f5...9d7c
2	Memory Image	Workstation	RAM capture (16 GB)	b7e1...4c2a
3	Event Logs	Windows Event Logs	Security, System, Application logs	c9d4...8f1e
4	Browser Data	Chrome	History, downloads, cookies	e2a1...7b3d
5	Email Archive	Outlook	PST file with relevant emails	f6c8...1a9e
6	Network Capture	Firewall	PCAP file (authentication log)	d4b9...6c0f

5 Methodology Section (What to Include)



- Forensic principles followed (e.g., NIST, ISO 27037).
- Tools used (e.g., Autopsy, FTK Imager, Wireshark).
- Step-by-step process for collection, analysis, and validation.
- How evidence integrity was maintained (e.g., write blockers, hash verification).
- Any assumptions made during the investigation.

6 Findings Section (Example)



- User account was accessed from an unfamiliar IP address (185.199.110.20) on 2025-02-10 at 19:42:10 UTC.
- Multiple failed login attempts were recorded before a successful login.
- Email settings were modified to forward messages to an external address.
- Suspicious files were downloaded from a malicious domain.

7 Limitations Section (Example)



- Some log files were not available (rotation/overwritten).
- Encrypted files could not be analysed.
- Network capture did not include full traffic.
- Timestamps may be affected by system clock differences.
- Analysis is based only on the evidence provided.

8 Recommendations (Example)



- Reset the affected user account and enforce multi-factor authentication (MFA).
- Review and monitor for further suspicious activity.
- Implement stronger log retention policies.
- Block malicious IP addresses and domains.
- Provide user awareness training on phishing risks.

Key Takeaways



- A good forensic report is clear, accurate, and objective.
- It tells a complete and factual story based on evidence.
- It should be easy to understand for both technical and non-technical readers.
- Always document tools, methods, and evidence details.
- Include limitations and provide practical recommendations.

Best Practices



- Use a consistent and professional structure.
- Be factual, unbiased, and evidence-driven.
- Include timelines, screenshots, and artifacts where relevant.
- Verify all findings and maintain evidence integrity.
- Write in clear and simple language.
- Ensure the report can stand up to internal or legal review.



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta

Your First Complete Forensic Investigation

A beginner capstone using authorized sample evidence, investigation steps, verification, and reporting.

1 Scenario Overview



- You are a junior analyst in a security team.
- A company suspects a user account was compromised, and suspicious activity was detected.
- You are provided with authorised sample evidence to investigate.
- Your goal is to analyse the evidence, identify what happened, and create a professional report.

2 Investigation Objectives



- Identify the initial point of compromise.
- Determine the attacker's actions.
- Analyse supporting evidence from different sources.
- Build a timeline of events.
- Verify all findings.
- Document the investigation in a clear and professional report.

3 Provided Sample Evidence (Example)

Evidence Type	Description	Example Files
 Windows Event Logs	System, Security, Application logs	EVTX files
 System Artifacts	Prefetch, Registry, Recycle Bin	.pf files, registry hives, \$Recycle.Bin
 Browser Data	History, downloads, cookies	Browser database files
 Email Evidence	Email headers and message files	.eml or .msg files
 Memory Image	Volatile memory data (optional)	.mem file
 Network Capture	Network traffic (optional)	.pcap file

4 Investigation Steps

- 1 Prepare the Environment**
Set up tools and verify the integrity of the sample evidence.
- 2 Review Initial Information**
Understand the scenario and identify key questions.
- 3 Analyse the Evidence**
Investigate logs, artifacts, browser data, email, memory and network captures.
- 4 Build a Timeline**
Correlate events from multiple sources.
- 5 Verify Findings**
Validate results and confirm with supporting evidence.
- 6 Create the Investigation Report**
Document scope, methodology, findings, and recommendations.

5 Tools You Will Use (Example)

Tool	Purpose	Use Case in This Investigation
 Autopsy	Disk analysis	Analyse disk image, recover files, view timelines
 FTK Imager	Acquire and verify images	Validate evidence integrity, create disk images
 Wireshark	Network analysis	Analyse network traffic (PCAP)
 Volatility 3	Memory analysis	Investigate memory image for malicious activity
 Event Viewer/Log Analysis	Log analysis	Review Windows event logs for suspicious activity
 Registry Explorer	Registry analysis	Analyse registry hives for persistence and activity

6 Key Findings (Example)



- User account was accessed from an unfamiliar IP address.
- Suspicious files were downloaded.
- Browser history shows access to malicious domain.
- Registry shows persistence mechanism.
- Evidence of data exfiltration attempt in network logs.

7 Report Output (Example)



- Executive Summary
- Scope and Objectives
- Methodology and Tools Used
- Evidence Inventory
- Detailed Findings (with screenshots)
- Timeline of Events
- Conclusion and Recommendations
- Limitations

8 Key Takeaways



- You can conduct a complete end-to-end forensic investigation using real tools and evidence.
- Multiple data sources provide a clearer and more accurate picture.
- A well-documented report is essential for incident response and legal cases.
- Always verify findings and maintain chain of custody.

9 Best Practices



- Use only authorised and legal sample evidence.
- Follow a structured investigation process.
- Document every step and maintain evidence integrity.
- Correlate multiple sources before making conclusions.
- Be clear, objective, and professional in your report.
- Include screenshots and timestamps for key findings.



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta



@veriqta